

Answer ALL questions.

Write your answers in the spaces provided.

- 1** Pieces of plant tissue, placed in a solution of salt (sodium chloride) in water, may gain or lose mass or remain unchanged.

(a) Explain the changes that occur when the mass increases.

(3)

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- (b) A student investigated the water potential of potato tissue by measuring the change in mass of pieces of potato in a range of concentrations of salt solutions.

The table shows the results of this investigation.

Concentration of salt solution / mol dm ⁻³	Mean percentage change in mass (%)
0.0	+17.0
0.2	+8.0
0.4	-6.0
0.6	-14.0
0.8	-20.5
1.0	-25.0

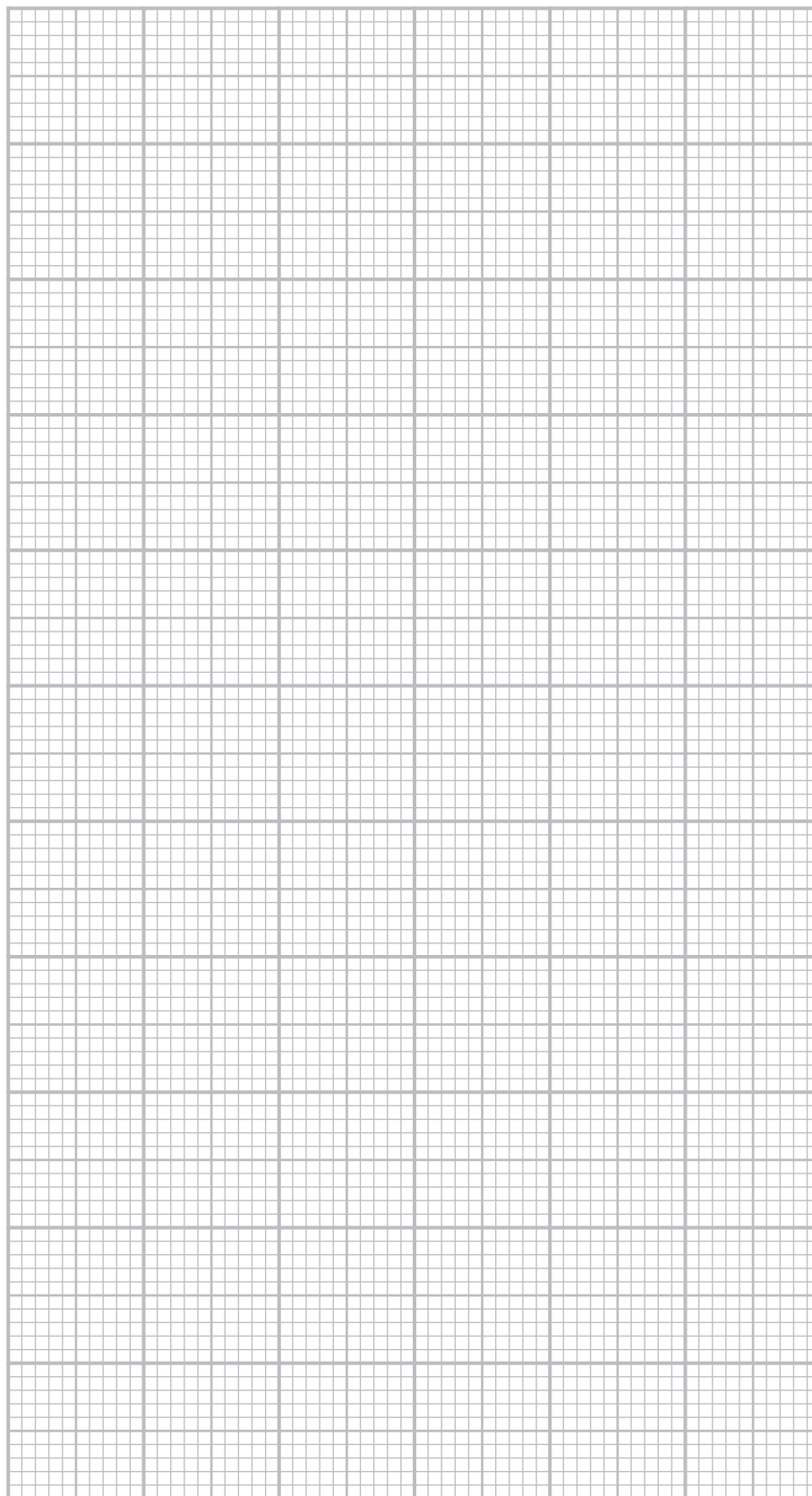
- (i) Describe a method that could have been used to obtain these results.

(5)

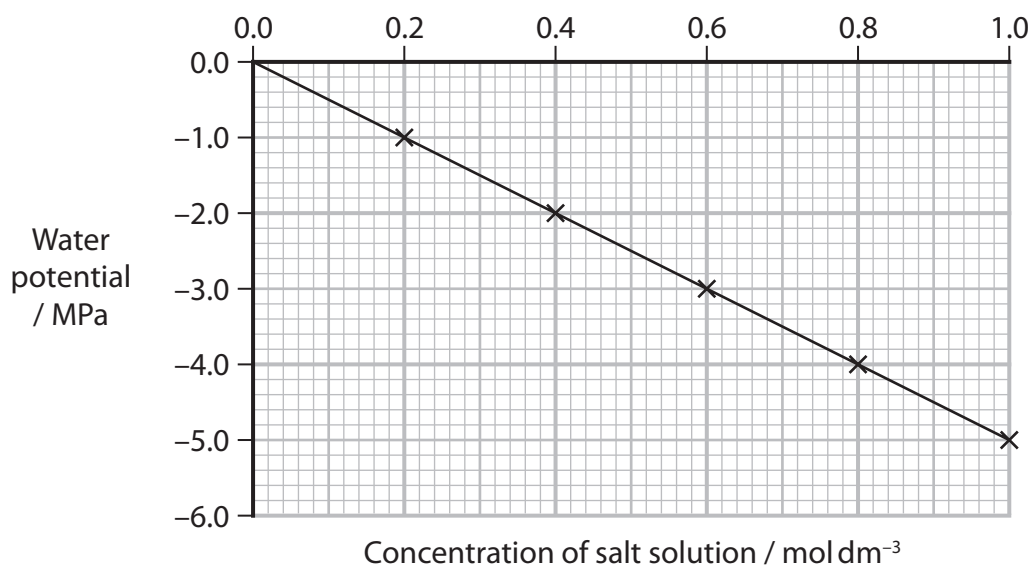


- (ii) Plot a graph of these results and draw a straight line of best fit through the points.

(4)



(iii) The graph shows the water potential of different concentrations of salt solution.



Determine the water potential of the potato tissue used in this investigation.

Indicate on each graph how you arrived at your answer.

(2)

Answer MPa

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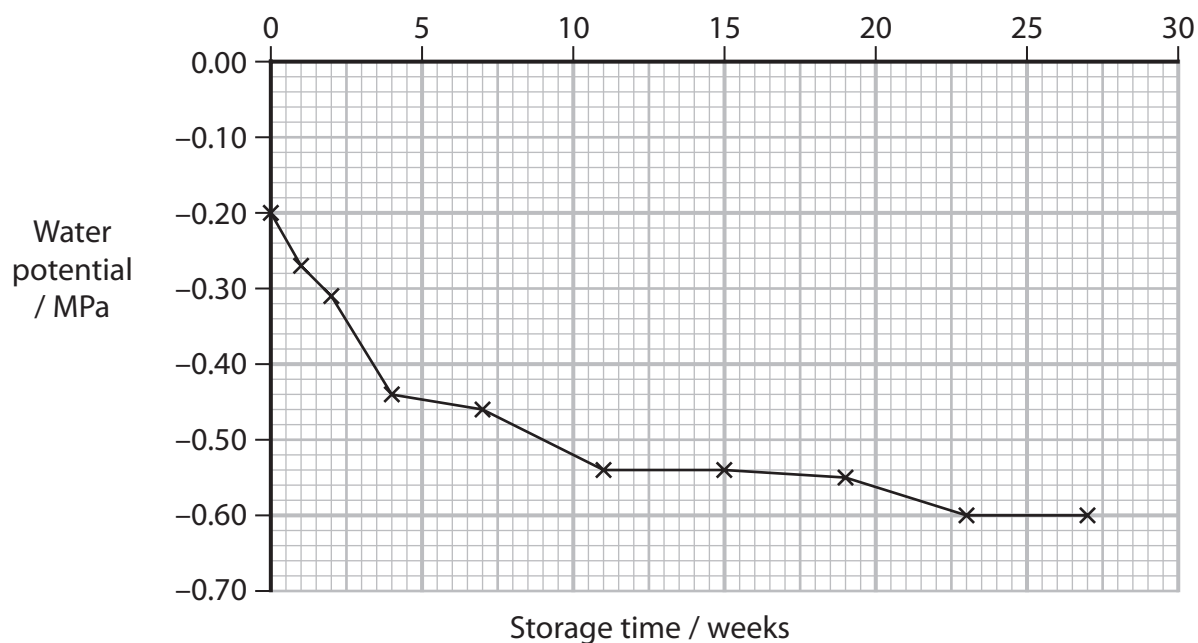
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- (c) A scientist investigated the effect of storage time on the water potential of potato tubers.

The results of this investigation are shown in the graph.



Explain these results.

(3)

(Total for Question 1 = 17 marks)

